

Non-Parametric Tests

Unit-3: Statistical Methods

Introduction to Non-Parametric Tests

Definition

Non-parametric tests are procedures for testing hypotheses about population distributions when

- the data are non-normal, or
- meaningful sample statistics (such as the mean and variance) cannot be calculated or are not appropriate.

They do not depend on population parameters such as the mean or variance. The chi-square test is a classical example of a non-parametric test. For non-normal data with sample size less than 50, we often resort to non-parametric tests.

Definition

Popular Non-Parametric Tests

- The Sign Test – Wilcoxon Signed-Rank Test
- The Rank Sum Test – Mann-Whitney U Test
- The One-Sample Runs Test
- The Kruskal-Wallis or H Test

The Sign Test (Wilcoxon Signed-Rank Test)

Definition

The **Sign Test** is one of the simplest non-parametric tests. It is based only on the number of plus (+) and minus (–) signs, ignoring the magnitudes of the observations.

Let S be the number of times the less frequent sign occurs (either + or –). For a sample of size n and a two-sided alternative at 5% level of significance (LOS), a commonly used approximate critical value is

$$K = \frac{n-1}{2} - 0.98\sqrt{n}.$$

If

$$S \leq K,$$

we reject the null hypothesis H_0 .

Types of Sign Test

Definition

Types of Sign Test

- **One-sample sign test:** We have a single sample of size n . Each observation is compared with a hypothesised median.
- **Paired-sample sign test:** Data are collected in pairs (e.g., before and after some treatment). Each pair (x_i, y_i) is replaced by
 - a plus sign + if $x_i > y_i$,
 - a minus sign – if $x_i < y_i$,
 - discarded if $x_i = y_i$.

Question

A typing school claims that in a six-week intensive course, it can train students to type, on the average, at least 60 words per minute. A random sample of 15 graduates is given a typing test, and the median number of words per minute typed by each of these students is as follows:

Student	A	B	C	D	E
Words per minute	81	76	53	71	66
Student	F	G	H	I	J
Words per minute	59	88	73	80	66
Student	K	L	M	N	O
Words per minute	58	70	60	56	55

Question

Test the hypothesis that the median typing speed of graduates is at least 60 words per minute at 5% LOS using the sign test.

Answer

We test whether the population median typing speed is at least 60 words per minute.

Step 1: Hypotheses.

$$H_0 : \text{Median typing speed } M_d = 60,$$

$$H_1 : \text{Median typing speed } M_d > 60.$$

Step 2: Assign signs. For each observation x_i we consider $x_i - 60$ and assign:

- + if $x_i > 60$,
- – if $x_i < 60$,

- discard if $x_i = 60$.

From the data,

$$\begin{aligned}\text{Number of } + \text{ signs} &= 9, \\ \text{Number of } - \text{ signs} &= 5, \\ \text{Number of zeros (ties)} &= 1.\end{aligned}$$

Hence, the effective sample size is

$$n = 9 + 5 = 14,$$

and the less frequent sign occurs

$$S = 5 \quad (\text{the number of } - \text{ signs}).$$

Step 3: Critical value. Using the approximate formula for a two-sided test at 5% LOS:

$$\begin{aligned}K &= \frac{n-1}{2} - 0.98\sqrt{n} \\ &= \frac{14-1}{2} - 0.98\sqrt{14} \\ &= 6.5 - 0.98\sqrt{14}.\end{aligned}$$

Since

$$\begin{aligned}\sqrt{14} &\approx 3.74, \\ K &\approx 6.5 - 0.98(3.74) \approx 6.5 - 3.67 \approx 2.83.\end{aligned}$$

Step 4: Decision. Here $S = 5 > K \approx 2.83$. Therefore, S is not small enough to reject H_0 .

We do not reject H_0 .

Final Answer

At the 5% level of significance, there is no evidence to contradict the typing school's claim. The median typing speed of graduates can be taken to be at least 60 words per minute.

Question

Use the sign test to determine whether there is a difference between the number of days until the collection of an account receivable before and after a new collection policy.

Before	30	28	34	35	40	42	33	38	34	45	28	27	25	41	36
After	32	29	33	32	37	43	40	41	37	44	27	33	30	38	36

Answer

We compare each pair (Before, After), taking

$$d_i = \text{Before}_i - \text{After}_i.$$

Assign + if $d_i > 0$, – if $d_i < 0$, and discard ties.

Counting the resulting signs gives:

$$\text{Number of + signs} = 6,$$

$$\text{Number of – signs} = 8,$$

$$\text{Number of ties} = 1.$$

Thus the effective sample size is

$$n = 6 + 8 = 14,$$

and

$$S = \min(6, 8) = 6.$$

As in the previous example, the approximate critical value is

$$K \approx 2.83.$$

Since $S = 6 > K$, the number of less frequent signs is not small enough to reject H_0 .

Therefore, we accept the null hypothesis that there is no significant difference in collection times before and after the new policy.

Final Answer

There is no significant difference between the collection times before and after the new policy, at the 5% level of significance.

Rank Sum Test: The Mann–Whitney U Test

Definition

The sign test for comparing two population distributions uses only the signs of the paired differences and ignores the magnitudes of the observations. This can discard useful information.

The **Mann–Whitney U test** is a rank sum test that partially avoids this loss by utilising the relative magnitudes of the observations. It helps us determine whether two samples have come from identical populations.

Definition

Suppose we have two independent samples of sizes n_1 and n_2 . Combine all $n_1 + n_2$

observations and assign ranks (from 1 to $n_1 + n_2$). Let

R_1 = sum of ranks of sample 1,

R_2 = sum of ranks of sample 2.

The Mann–Whitney statistics are

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1,$$

$$U_2 = n_1 n_2 - U_1,$$

$$U = \min(U_1, U_2).$$

For large samples, U is approximately normal with

$$\mu_U = \frac{n_1 n_2}{2},$$

$$\sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}},$$

and the test statistic

$$Z = \frac{U - \mu_U}{\sigma_U}.$$

If $|Z|$ is less than the critical value from the standard normal distribution, we accept H_0 .

Question

Twenty-three applicants for a position are interviewed by three administrators and rated on a 5-point scale. The sum of the three ratings gives a “suitability score”. Although college education is not a requirement, the personnel director suspects that it might affect suitability. The raters do not know the educational background of the applicants.

Twelve applicants have completed at least two years of college.

Use the Mann–Whitney U test at 5% LOS to determine whether there is a difference in scores between:

- Group A: Applicants with less than two years of college,
- Group B: Applicants with at least two years of college.

Suitability scores are:

Group A	Group B
7	8
11	9
9	13
4	14
8	11
6	10
12	12
11	14
9	13
10	9
11	10
	8

Answer**Step 1: Hypotheses.**

H_0 : The two groups come from identical populations,

H_1 : The two groups do not come from identical populations.

Step 2: Ranks. Combine the 23 scores and assign ranks from 1 (smallest) to 23 (largest), handling ties by assigning average ranks. Let R_1 be the sum of ranks for Group A and R_2 for Group B.

From the ranking table (see original solution),

$$n_1 = 11, \quad n_2 = 12,$$

$$R_1 = 105, \quad R_2 = 171.$$

Step 3: Compute U_1, U_2, U .

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1$$

$$= 11 \times 12 + \frac{11 \times 12}{2} - 105$$

$$= 132 + 66 - 105$$

$$= 93,$$

$$U_2 = n_1 n_2 - U_1$$

$$= 132 - 93$$

$$= 39.$$

Thus

$$U = \min(U_1, U_2) = 39.$$

Step 4: Mean and standard deviation of U .

$$\begin{aligned}\mu_U &= \frac{n_1 n_2}{2} = \frac{11 \times 12}{2} = 66, \\ \sigma_U &= \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}} \\ &= \sqrt{\frac{11 \times 12 \times 24}{12}} \\ &= \sqrt{264} \approx 16.25.\end{aligned}$$

Step 5: Test statistic.

$$Z = \frac{U - \mu_U}{\sigma_U} = \frac{39 - 66}{16.25} \approx \frac{-27}{16.25} \approx -1.66.$$

At 5% LOS for a two-sided test,

$$z_{0.025} = 1.96.$$

Since

$$|Z| \approx 1.66 < 1.96,$$

we do not reject H_0 .

Final Answer

At the 5% level of significance, there is no significant difference between the suitability scores of the two groups. Educational background (less than two years vs. at least two years of college) does not appear to affect the suitability scores.

One-Sample Runs Test

Definition

The **one-sample runs test** is used to judge the randomness of a sequence based on the order in which observations occur.

A *run* is a succession of identical symbols that is preceded and followed by a different symbol (or by no symbol at all).

Let:

- n_1 and n_2 be the numbers of two types of symbols (e.g. M and W),
- r be the total number of runs.

For large samples (typically $n_1 > 20$ or $n_2 > 20$), r is approximately normal with

$$\begin{aligned}\mu_r &= \frac{2n_1 n_2}{n_1 + n_2} + 1, \\ \sigma_r^2 &= \frac{2n_1 n_2 (2n_1 n_2 - n_1 - n_2)}{(n_1 + n_2)^2 (n_1 + n_2 - 1)},\end{aligned}$$

and test statistic

$$Z = \frac{r - \mu_r}{\sigma_r}.$$

Question

An arrangement of 25 men (M) and 15 women (W) are lined up to purchase tickets for a premier show. Using the observed arrangement, the number of runs is found to be $r = 17$.

Test whether the arrangement is random at the 5% LOS.

Answer

We have

$$n_1 = 25, \quad n_2 = 15, \quad r = 17.$$

Mean of r .

$$\begin{aligned} \mu_r &= \frac{2n_1n_2}{n_1 + n_2} + 1 \\ &= \frac{2 \times 25 \times 15}{40} + 1 \\ &= \frac{750}{40} + 1 \\ &= 18.75 + 1 \\ &= 19.75. \end{aligned}$$

Variance of r .

$$\begin{aligned} \sigma_r^2 &= \frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2(n_1 + n_2 - 1)} \\ &= \frac{2 \times 25 \times 15(2 \times 25 \times 15 - 25 - 15)}{40^2 \times 39} \\ &= \frac{750 \times (750 - 40)}{1600 \times 39} \\ &= \frac{750 \times 710}{1600 \times 39} \\ &\approx 8.526, \\ \sigma_r &\approx \sqrt{8.526} \approx 2.92. \end{aligned}$$

Test statistic.

$$Z = \frac{r - \mu_r}{\sigma_r} = \frac{17 - 19.75}{2.92} \approx \frac{-2.75}{2.92} \approx -0.94.$$

At 5% LOS for a two-sided test,

$$z_{0.025} = 1.96.$$

Since

$$|Z| \approx 0.94 < 1.96,$$

we do not reject the null hypothesis of randomness.

Final Answer

There is no evidence at the 5% LOS to suggest that the arrangement is not random.

The Kruskal–Wallis or H Test

Definition

The **Kruskal–Wallis H test** is a non-parametric alternative to one-way analysis of variance (ANOVA). It is used to test whether several samples (three or more) come from the same population when normality or equal variances cannot be assumed. All observations from the k groups are combined and ranked from lowest to highest as if they were one sample.

Definition

Let:

- k be the number of groups,
- n_i the size of group i , for $i = 1, \dots, k$,
- $N = n_1 + n_2 + \dots + n_k$,
- R_i the sum of ranks for group i .

The Kruskal–Wallis statistic is

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1).$$

For sufficiently large samples, H is approximately chi-square distributed with $k - 1$ degrees of freedom.

Question

A company's trainees are randomly assigned to groups and taught an industrial inspection procedure by three different methods. At the end of the training, they are tested for inspection performance quality.

Scores:

Method A: 80, 83, 79, 85, 90, 68,

Method B: 82, 84, 60, 72, 86, 67, 91,

Method C: 93, 65, 77, 78, 88.

Use the Kruskal–Wallis H test at the 5% LOS to determine whether the three

methods are equally effective.

Answer

Step 1: Hypotheses.

H_0 : All three methods are equally effective (same distribution).

H_1 : At least one method differs in effectiveness.

Step 2: Ranks. Combine all 18 observations and assign ranks from 1 (smallest) to 18 (largest). Let R_1 , R_2 , and R_3 be the sums of ranks for Methods A, B, and C respectively.

From the rank table, we obtain

$$\begin{aligned} n_1 &= 6, & n_2 &= 7, & n_3 &= 5, \\ N &= 18, \\ R_1 &= 61, & R_2 &= 62, & R_3 &= 48. \end{aligned}$$

Step 3: Compute H .

$$\begin{aligned} H &= \frac{12}{N(N+1)} \left(\frac{R_1^2}{n_1} + \frac{R_2^2}{n_2} + \frac{R_3^2}{n_3} \right) - 3(N+1) \\ &= \frac{12}{18 \times 19} \left(\frac{61^2}{6} + \frac{62^2}{7} + \frac{48^2}{5} \right) - 3 \times 19. \end{aligned}$$

Numerically,

$$\begin{aligned} \frac{61^2}{6} &\approx 620.17, \\ \frac{62^2}{7} &\approx 549.14, \\ \frac{48^2}{5} &\approx 460.80. \end{aligned}$$

Thus

$$\begin{aligned} H &\approx \frac{12}{342} (620.17 + 549.14 + 460.80) - 57 \\ &\approx \frac{12}{342} (1630.11) - 57 \\ &\approx 57.20 - 57 \\ &\approx 0.20. \end{aligned}$$

Step 4: Critical value and decision. Degrees of freedom:

$$v = k - 1 = 3 - 1 = 2.$$

From chi-square tables,

$$\chi_{0.05,2}^2 = 5.991.$$

Since

$$H \approx 0.20 < 5.991,$$

we do not reject H_0 .

Final Answer

At the 5% level of significance, the Kruskal–Wallis test indicates no significant difference among the three methods. The methods may be regarded as equally effective.

Question Bank with Solutions

Two Marks Questions

Question**Question 1 (2 marks)**

What are the main assumptions required for the Mann–Whitney U -test?

Answer

The Mann–Whitney U -test requires:

- Independent random samples from two populations.
- Observations are ordinal or continuous.
- The two populations have similar shapes (for comparing locations such as medians).

Question**Question 2 (2 marks)**

State the null and alternative hypotheses for the Wilcoxon signed-rank test.

Answer

Let d denote the paired difference (e.g. After – Before).

H_0 : Median difference is zero (no significant difference),

H_1 : Median difference is not zero (a significant difference exists).

Alternatively, the hypotheses may be written as

$$H_0 : \mu_d = 0,$$

$$H_1 : \mu_d \neq 0,$$

where μ_d denotes the median (or mean, in some contexts) of the differences.

Question**Question 3 (2 marks)**

What does a “run” represent in the runs test for randomness?

Answer

A **run** is a sequence of consecutive identical symbols or values in a series. For example, in the sequence

A A A B B A

there are 4 runs: AAA, BB, and A at the end. The runs test examines whether the observed number of runs is compatible with randomness or indicates a pattern in the data sequence.

Question**Question 4 (2 marks)**

When is the Kruskal–Wallis H -test preferred over one-way ANOVA?

Answer

The Kruskal–Wallis H -test is preferred when:

- Data are not normally distributed.
- Variances are unequal across groups.
- Data are ordinal rather than interval/ratio.
- Sample sizes are small so that normality assumptions are doubtful.

Question**Question 5 (2 marks)**

Calculate the Mann–Whitney U statistic for the following ranks from Group 1: 2, 4, 6, 8 ($n_1 = 4$) and Group 2: 1, 3, 5, 7, 9, 10 ($n_2 = 6$).

Answer

First compute the sum of ranks for Group 1:

$$R_1 = 2 + 4 + 6 + 8 = 20.$$

The Mann–Whitney statistic U_1 is

$$\begin{aligned} U_1 &= R_1 - \frac{n_1(n_1 + 1)}{2} \\ &= 20 - \frac{4 \cdot 5}{2} \\ &= 20 - 10 \\ &= 10. \end{aligned}$$

For Group 2,

$$\begin{aligned}U_2 &= n_1 n_2 - U_1 \\&= 4 \times 6 - 10 \\&= 24 - 10 \\&= 14.\end{aligned}$$

Hence, the test statistic is

$$U = \min(U_1, U_2) = \min(10, 14) = 10.$$

Question

Question 6 (2 marks)

What is the test statistic formula for the Wilcoxon signed-rank test?

Answer

Let T^+ be the sum of ranks of positive differences and T^- the sum of ranks of negative differences. The Wilcoxon signed-rank test statistic is

$$T = \min(T^+, T^-).$$

For large samples, the standardized statistic is

$$Z = \frac{T - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}},$$

where n is the number of non-zero differences.

Question

Question 7 (2 marks)

In a runs test, if we have 15 runs with $n_1 = 12$ and $n_2 = 8$, calculate the expected number of runs.

Answer

The expected number of runs is

$$\begin{aligned}
 E(R) &= \frac{2n_1n_2}{n_1 + n_2} + 1 \\
 &= \frac{2 \cdot 12 \cdot 8}{12 + 8} + 1 \\
 &= \frac{192}{20} + 1 \\
 &= 9.6 + 1 \\
 &= 10.6.
 \end{aligned}$$

Question**Question 8 (2 marks)**

What is the chi-square approximation formula for the Kruskal–Wallis H -test?

Answer

Let

- N be the total sample size,
- k be the number of groups,
- R_i be the sum of ranks for group i ,
- n_i be the sample size of group i .

The Kruskal–Wallis statistic is

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1).$$

For sufficiently large samples, H approximately follows a chi-square distribution with $(k-1)$ degrees of freedom.

Question**Question 9 (2 marks)**

What are the conditions under which the normal approximation can be used for the Mann–Whitney U -test?

Answer

The normal approximation for the Mann–Whitney U -test may be used when

- both $n_1 \geq 8$ and $n_2 \geq 8$, or
- the smaller sample size is at least 3 and the product $n_1n_2 \geq 20$.

Then the standardized statistic is

$$Z = \frac{U - \mu_U}{\sigma_U},$$

where

$$\mu_U = \frac{n_1 n_2}{2},$$

$$\sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}.$$

Question

Question 10 (2 marks)

State the decision rule for the runs test when testing for randomness at $\alpha = 0.05$.

Answer

For the runs test:

- If the observed number of runs is significantly low or significantly high compared with the expected number of runs, reject H_0 (randomness).
- For large samples, compute

$$Z = \frac{R - E(R)}{\sigma_R},$$

and reject H_0 if

$$|Z| > 1.96 \quad (\alpha = 0.05, \text{ two-tailed}).$$

- Too few runs suggest clustering; too many runs suggest systematic alternation.

Ten Marks Questions

Question

Question 11 (10 marks)

Two different teaching methods are compared by testing students' performance scores. Use the Mann–Whitney U -test to determine if there is a significant difference between the two methods at $\alpha = 0.05$.

Method A: 78, 85, 92, 88, 91, 87, 94

Method B: 82, 76, 89, 84, 86, 90, 83, 79.

Answer**Step 1: State hypotheses**

H_0 : No difference in median performance between methods,

H_1 : A significant difference exists between methods.

Step 2: Combine and rank all observations

Combined data (ascending):

76, 78, 79, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 94.

Value	Rank	Method
76	1	B
78	2	A
79	3	B
82	4	B
83	5	B
84	6	B
85	7	A
86	8	B
87	9	A
88	10	A
89	11	B
90	12	B
91	13	A
92	14	A
94	15	A

Step 3: Calculate rank sums

For Method A:

$$R_A = 2 + 7 + 9 + 10 + 13 + 14 + 15 = 70.$$

For Method B:

$$R_B = 1 + 3 + 4 + 5 + 6 + 8 + 11 + 12 = 50.$$

Step 4: Calculate U statistics

Here $n_A = 7$, $n_B = 8$.

$$U_A = R_A - \frac{n_A(n_A + 1)}{2} = 70 - \frac{7 \cdot 8}{2} = 70 - 28 = 42,$$

$$U_B = R_B - \frac{n_B(n_B + 1)}{2} = 50 - \frac{8 \cdot 9}{2} = 50 - 36 = 14.$$

Check:

$$U_A + U_B = 42 + 14 = 56 = n_A n_B = 7 \times 8.$$

Step 5: Test statistic

$$U = \min(U_A, U_B) = \min(42, 14) = 14.$$

Step 6: Critical value

For $n_A = 7$, $n_B = 8$ at $\alpha = 0.05$ (two-tailed), the critical value from Mann–Whitney tables is $U_{0.05} = 8$.

Step 7: Decision

Since

$$U = 14 > 8,$$

we fail to reject H_0 .

Final Answer

There is no significant difference between the two teaching methods at the 5% significance level.

Question**Question 12 (10 marks)**

Ten subjects were measured before and after a training program. Use the Wilcoxon signed-rank test to determine if the training program is effective at $\alpha = 0.05$.

Subject	1	2	3	4	5	6	7	8	9	10
Before	65	78	82	74	69	85	71	79	77	73
After	72	81	85	78	75	87	76	84	80	76

Answer**Step 1: Hypotheses**

H_0 : Median difference = 0 (no improvement),

H_1 : Median difference > 0 (improvement exists).

Step 2: Differences and ranks

Subject	Before	After	Difference d	$ d $	Rank
1	65	72	+7	7	8.5
2	78	81	+3	3	3.5
3	82	85	+3	3	3.5
4	74	78	+4	4	5.5
5	69	75	+6	6	7
6	85	87	+2	2	1.5
7	71	76	+5	5	6
8	79	84	+5	5	6
9	77	80	+3	3	3.5
10	73	76	+3	3	3.5

(Ties in absolute differences receive average ranks.)

Step 3: Rank sums

All differences are positive, so

$$\begin{aligned}
 T^+ &= 8.5 + 3.5 + 3.5 + 5.5 + 7 + 1.5 + 6 + 6 + 3.5 + 3.5 \\
 &= 48.5, \\
 T^- &= 0.
 \end{aligned}$$

Thus

$$T = \min(T^+, T^-) = \min(48.5, 0) = 0.$$

Step 5: Critical value

For $n = 10$ non-zero pairs, at $\alpha = 0.05$ (one-tailed), the critical value from Wilcoxon tables is $T_{0.05} = 10$.

Step 6: Decision

Since

$$T = 0 < 10,$$

we reject H_0 .

Final Answer

The training program is significantly effective at improving performance at $\alpha = 0.05$.

Question

Question 13 (10 marks)

A quality control manager wants to test if a production process is random. The following sequence represents defective (D) and non-defective (N) items in order of production. Test for randomness at $\alpha = 0.05$.

Sequence:

N N N D D N N D D D N N N D N D D N N N D D D N N

Answer**Step 1: Hypotheses**

H_0 : The sequence is random,

H_1 : The sequence is not random.

Step 2: Count elements and runs

Number of non-defectives (N):

$$n_1 = 15;$$

number of defectives (D):

$$n_2 = 10;$$

total observations:

$$N = n_1 + n_2 = 25.$$

Identify runs in the sequence (separating each run by “|”):

N N N | D D | N N | D D D | N N N | D | N | D D | N N N | D D D | N N.

Number of runs:

$$R = 11.$$

Step 3: Expected number of runs

$$\begin{aligned} E(R) &= \frac{2n_1n_2}{n_1 + n_2} + 1 \\ &= \frac{2 \cdot 15 \cdot 10}{15 + 10} + 1 \\ &= \frac{300}{25} + 1 \\ &= 12 + 1 = 13. \end{aligned}$$

Step 4: Standard deviation of runs

$$\begin{aligned} \sigma_R &= \sqrt{\frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2(n_1 + n_2 - 1)}} \\ &= \sqrt{\frac{2 \cdot 15 \cdot 10(2 \cdot 15 \cdot 10 - 15 - 10)}{25^2 \cdot 24}} \\ &= \sqrt{\frac{300 \cdot 275}{625 \cdot 24}} = \sqrt{\frac{82500}{15000}} = \sqrt{5.5} \approx 2.345. \end{aligned}$$

Step 5: Test statistic

$$Z = \frac{R - E(R)}{\sigma_R} = \frac{11 - 13}{2.345} = \frac{-2}{2.345} \approx -0.853.$$

Step 6: Critical values

For $\alpha = 0.05$ (two-tailed), the critical Z -values are

$$Z_{0.025} = \pm 1.96.$$

Step 7: Decision

Since

$$|Z| = 0.853 < 1.96,$$

we fail to reject H_0 .

Final Answer

There is insufficient evidence to conclude that the production process is non-random. The sequence appears to be random at the 5% significance level.

Question**Question 14 (10 marks)**

Three different fertilizers are tested on crop yields. Use the Kruskal–Wallis H -test to determine if there is a significant difference between the fertilizers at $\alpha = 0.05$.

Fertilizer A: 45, 52, 48, 51, 46

Fertilizer B: 41, 38, 44, 42, 39, 43

Fertilizer C: 55, 58, 53, 56, 60.

Answer**Step 1: Hypotheses**

H_0 : No difference in median yields between fertilizers,

H_1 : At least one fertilizer has a different median yield.

Step 2: Combine and rank all observations

Combined ordered data:

38, 39, 41, 42, 43, 44, 45, 46, 48, 51, 52, 53, 55, 56, 58, 60.

Value	Rank	Fertilizer	Group
38	1	B	B
39	2	B	B
41	3	B	B
42	4	B	B
43	5	B	B
44	6	B	B
45	7	A	A
46	8	A	A
48	9	A	A
51	10	A	A
52	11	A	A
53	12	C	C
55	13	C	C
56	14	C	C
58	15	C	C
60	16	C	C

Step 3: Rank sums and sample sizes

$$R_A = 7 + 8 + 9 + 10 + 11 = 45,$$

$$R_B = 1 + 2 + 3 + 4 + 5 + 6 = 21,$$

$$R_C = 12 + 13 + 14 + 15 + 16 = 70.$$

Sample sizes:

$$n_A = 5, \quad n_B = 6, \quad n_C = 5,$$

$$N = 16.$$

Step 4: Calculate H -statistic

$$\begin{aligned}
 H &= \frac{12}{N(N+1)} \left(\frac{R_A^2}{n_A} + \frac{R_B^2}{n_B} + \frac{R_C^2}{n_C} \right) - 3(N+1) \\
 &= \frac{12}{16 \cdot 17} \left(\frac{45^2}{5} + \frac{21^2}{6} + \frac{70^2}{5} \right) - 3 \cdot 17.
 \end{aligned}$$

Compute each term:

$$\frac{45^2}{5} = \frac{2025}{5} = 405,$$

$$\frac{21^2}{6} = \frac{441}{6} = 73.5,$$

$$\frac{70^2}{5} = \frac{4900}{5} = 980.$$

Thus

$$\begin{aligned} H &= \frac{12}{272}(405 + 73.5 + 980) - 51 \\ &= \frac{12}{272}(1458.5) - 51 \\ &= \frac{17502}{272} - 51 \\ &\approx 64.35 - 51 \\ &\approx 13.35. \end{aligned}$$

Step 5: Critical value

Degrees of freedom:

$$df = k - 1 = 3 - 1 = 2.$$

For $\alpha = 0.05$ and $df = 2$,

$$\chi_{0.05,2}^2 = 5.991.$$

Step 6: Decision

Since

$$H = 13.35 > 5.991,$$

we reject H_0 .

Final Answer

There is a significant difference between the fertilizers' effects on crop yields at the 5% significance level. Post-hoc analysis indicates that Fertilizer C appears to be the most effective, while Fertilizer B shows the lowest yields.

One-Way ANOVA (Analysis of Variance)

Notes and Worked Examples

Introduction to ANOVA

Definition

ANOVA (Analysis of Variance) is a statistical method used to analyse the differences among group means in a sample. It helps determine whether there are any statistically significant differences between the means of three or more independent (unrelated) groups.

ANOVA works by comparing the variation within groups (due to random error) to the variation between groups (due to treatment effects). If the between-group variation is significantly larger than the within-group variation, we conclude that at least one group mean is different from the others.

Overview

There are primarily four widely used designs in statistical experimentation. Each of these designs controls variation differently and applies to different research settings.

Four Types of Experimental Designs

1. Completely Randomized Design (CRD) – 1-way ANOVA
2. Randomized Block Design (RBD) – 2-way ANOVA
3. Latin Square Design (LSD) – 3-way ANOVA
4. 2^2 Factorial Design – 2-way ANOVA (balanced factorial structure)

Topic

Completely Randomized Design (CRD)

Definition

A **Completely Randomized Design (CRD)** is the simplest form of experimental design where experimental units are randomly assigned to treatments without any blocking or classification structure. It is appropriate when all experimental units are homogeneous.

Characteristics:

- Random assignment of all treatments.

- Assumes uniformity among experimental units.
- Easy to implement and analyze.
- Does not control variability across unit differences. :contentReference[oaicite:6]index=6

Topic

Randomized Block Design (RBD) :contentReference[oaicite:7]index=7

Definition

A **Randomized Block Design (RBD)** is used when experimental units differ systematically. Units are grouped into **blocks** based on a factor expected to influence the outcome, and treatments are randomly assigned within each block. This reduces block-level variability. :contentReference[oaicite:8]index=8

Characteristics:

- Accounts for natural variation by grouping similar units.
- Treatments randomized within blocks.
- More precise results when variability among blocks exists.
- Reduces experimental error substantially. :contentReference[oaicite:9]index=9

Topic

Latin Square Design (LSD) :contentReference[oaicite:10]index=10

Definition

A **Latin Square Design (LSD)** is applied when there are two external sources of variation (e.g., rows and columns). Each treatment appears exactly once per row and once per column, controlling both variation sources. :contentReference[oaicite:11]index=11

Characteristics:

- Controls variability across two blocking factors.
- Efficient with limited resources.
- Treatment appears once in each row and column.
- Used when three factors affect response (two nuisance + treatment). :contentReference[oaicite:12]index=12

Topic

2² Factorial Design :contentReference[oaicite:13]index=13

Definition

A 2² **Factorial Design** studies two independent factors, each at two levels (e.g., High–Low). It allows estimation of both main effects and interaction effects. Total treatment combinations = $2 \times 2 = 4$. :contentReference[oaicite:14]index=14

Characteristics:

- Two factors with two levels each.
- Fully crossed design with four treatment combinations.
- Measures both individual and joint effects.
- Highly efficient and widely applicable. :contentReference[oaicite:15]index=15

Classifications of ANOVA

Definition

1. One-Way ANOVA (Single Factor ANOVA)

Compares the means of three or more groups based on a single independent variable (or factor).

Example: Testing the effect of different diets on weight loss, where the independent variable is *diet type* (with three or more categories).

Definition

2. Two-Way ANOVA (Factorial ANOVA)

Compares the means of groups based on two independent variables (factors), assessing both

- the main effect of each factor, and
- the interaction effect between the factors.

Example: Testing the effect of different diets and exercise routines on weight loss, where the independent variables are *diet type* and *exercise routine*.

Definition

3. MANOVA (Multivariate ANOVA)

Extends ANOVA to multiple dependent variables, analysing the relationship between them and the independent variable(s).

Example: Testing the effect of a treatment on multiple health outcomes (e.g., blood

pressure, cholesterol, and heart rate).

One-Way ANOVA: Hypotheses and Formulae

Definition

Hypotheses for One-Way ANOVA

Consider k groups with means $\mu_1, \mu_2, \dots, \mu_k$.

- **Null Hypothesis H_0 :** All group means are equal

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k.$$

- **Alternative Hypothesis H_1 :** At least one group mean is different from the others.

Definition

F-statistic

The test statistic for one-way ANOVA is

$$F = \frac{\text{MSB}}{\text{MSW}},$$

where

$$\text{MSB} = \text{Mean Square Between groups} = \frac{\text{SSB}}{k - 1},$$

$$\text{MSW} = \text{Mean Square Within groups} = \frac{\text{SSW}}{N - k},$$

N is the total number of observations and k is the number of groups.

If the calculated F value is greater than the critical value from the F -distribution table at a chosen significance level α , we reject the null hypothesis.

Definition

Sums of Squares for One-Way ANOVA

Let x_{ij} denote the observation from the i -th treatment (group) and j -th replication, T_i be the total of the i -th treatment, T the grand total, and n_i the number of observations in group i .

1. Total Sum of Squares (SST):

$$\text{SST} = \sum_i \sum_j x_{ij}^2 - \frac{T^2}{N}.$$

This represents the total variation in the data.

2. Sum of Squares Between Groups (SSB):

$$SSB = \sum_i \frac{T_i^2}{n_i} - \frac{T^2}{N}.$$

This measures the variation between the group means.

3. Sum of Squares Within Groups (SSW):

$$SSW = SST - SSB.$$

This is the variation within each group.

Worked Example 1: Diet Plans**Question**

You want to test if there is a significant difference in the mean weight loss for three different diet plans A, B, and C. The data collected (in kg) are as follows:

Diet A: 3, 4, 5, 6, 2,

Diet B: 8, 9, 7, 10, 9,

Diet C: 5, 6, 7, 8, 7.

Test at 5% level of significance using one-way ANOVA.

Answer**Step 1: State the hypotheses**

$$H_0 : \mu_A = \mu_B = \mu_C \quad (\text{all means are equal}),$$

$$H_1 : \text{At least one mean is different.}$$

Step 2: Calculate T and N

Grand total:

$$\begin{aligned} T &= 3 + 4 + 5 + 6 + 2 + 8 + 9 + 7 + 10 + 9 + 5 + 6 + 7 + 8 + 7 \\ &= 96. \end{aligned}$$

Total number of observations:

$$N = 15.$$

Step 3: Calculate $\sum \sum x_{ij}^2$

For Diet A:

$$3^2 + 4^2 + 5^2 + 6^2 + 2^2 = 9 + 16 + 25 + 36 + 4 = 90.$$

For Diet B:

$$8^2 + 9^2 + 7^2 + 10^2 + 9^2 = 64 + 81 + 49 + 100 + 81 = 375.$$

For Diet C:

$$5^2 + 6^2 + 7^2 + 8^2 + 7^2 = 25 + 36 + 49 + 64 + 49 = 223.$$

Hence

$$\sum \sum x_{ij}^2 = 90 + 375 + 223 = 688.$$

Step 4: Calculate SST

$$\begin{aligned} \text{SST} &= \sum \sum x_{ij}^2 - \frac{T^2}{N} \\ &= 688 - \frac{96^2}{15} \\ &= 688 - \frac{9216}{15} \\ &= 688 - 614.4 \\ &= 73.6. \end{aligned}$$

Step 5: Calculate SSB

Group totals:

$$\begin{aligned} T_A &= 3 + 4 + 5 + 6 + 2 = 20, \\ T_B &= 8 + 9 + 7 + 10 + 9 = 43, \\ T_C &= 5 + 6 + 7 + 8 + 7 = 33. \end{aligned}$$

Group sizes:

$$n_A = n_B = n_C = 5.$$

$$\begin{aligned} \text{SSB} &= \frac{T_A^2}{n_A} + \frac{T_B^2}{n_B} + \frac{T_C^2}{n_C} - \frac{T^2}{N} \\ &= \frac{20^2}{5} + \frac{43^2}{5} + \frac{33^2}{5} - \frac{96^2}{15}. \end{aligned}$$

Compute each term:

$$\begin{aligned} \frac{20^2}{5} &= \frac{400}{5} = 80, \\ \frac{43^2}{5} &= \frac{1849}{5} = 369.8, \\ \frac{33^2}{5} &= \frac{1089}{5} = 217.8. \end{aligned}$$

Thus

$$\begin{aligned} \text{SSB} &= 80 + 369.8 + 217.8 - 614.4 \\ &= 667.6 - 614.4 \\ &= 54.4. \end{aligned}$$

Step 6: Calculate SSW

$$\begin{aligned}
 \text{SSW} &= \text{SST} - \text{SSB} \\
 &= 73.6 - 54.4 \\
 &= 19.2.
 \end{aligned}$$

Step 7: Mean squares and F -statistic

Here $k = 3$ groups, $N = 15$.

$$\begin{aligned}
 \text{MSB} &= \frac{\text{SSB}}{k - 1} = \frac{54.4}{2} = 27.2, \\
 \text{MSW} &= \frac{\text{SSW}}{N - k} = \frac{19.2}{12} = 1.6.
 \end{aligned}$$

$$F = \frac{\text{MSB}}{\text{MSW}} = \frac{27.2}{1.6} = 17.$$

Step 8: Decision

Degrees of freedom:

$$\begin{aligned}
 v_1 &= k - 1 = 2, \\
 v_2 &= N - k = 12.
 \end{aligned}$$

From F -tables at 5% LOS,

$$F_{0.05}(2, 12) = 3.89.$$

Since

$$F = 17 > 3.89,$$

we reject H_0 .

Final Answer

There is a significant difference in the mean weight loss among the three diet plans at the 5% level of significance. At least one diet is significantly different from the others.

Worked Example 2: Sales in Three Regions**Question**

Sales data for three regions (North, South, West) are:

North: 20, 21, 22, 23, 19,
 South: 25, 27, 28, 26, 24,
 West: 18, 20, 19, 21, 22.

Test whether average sales differ between the three regions using one-way ANOVA

at 5% LOS.

Answer

Step 1: Hypotheses

$$H_0 : \mu_{\text{North}} = \mu_{\text{South}} = \mu_{\text{West}},$$

H_1 : At least one mean is different.

Step 2: Calculate T and N

$$\begin{aligned} T &= 20 + 21 + 22 + 23 + 19 + 25 + 27 + 28 + 26 + 24 \\ &\quad + 18 + 20 + 19 + 21 + 22 \\ &= 335, \\ N &= 15. \end{aligned}$$

Step 3: Calculate $\sum \sum x_{ij}^2$

For North:

$$20^2 + 21^2 + 22^2 + 23^2 + 19^2 = 400 + 441 + 484 + 529 + 361 = 2215.$$

For South:

$$25^2 + 27^2 + 28^2 + 26^2 + 24^2 = 625 + 729 + 784 + 676 + 576 = 3390.$$

For West:

$$18^2 + 20^2 + 19^2 + 21^2 + 22^2 = 324 + 400 + 361 + 441 + 484 = 2010.$$

Hence

$$\sum \sum x_{ij}^2 = 2215 + 3390 + 2010 = 7615.$$

Step 4: Calculate SST

$$\begin{aligned} \text{SST} &= \sum \sum x_{ij}^2 - \frac{T^2}{N} \\ &= 7615 - \frac{335^2}{15} \\ &= 7615 - \frac{112225}{15} \\ &= 7615 - 7489 \\ &= 126. \end{aligned}$$

Step 5: Calculate SSB

Group totals:

$$T_{\text{North}} = 20 + 21 + 22 + 23 + 19 = 105,$$

$$T_{\text{South}} = 25 + 27 + 28 + 26 + 24 = 130,$$

$$T_{\text{West}} = 18 + 20 + 19 + 21 + 22 = 100.$$

Group sizes:

$$n_{\text{North}} = n_{\text{South}} = n_{\text{West}} = 5.$$

$$\begin{aligned} \text{SSB} &= \frac{T_{\text{North}}^2}{n_{\text{North}}} + \frac{T_{\text{South}}^2}{n_{\text{South}}} + \frac{T_{\text{West}}^2}{n_{\text{West}}} - \frac{T^2}{N} \\ &= \frac{105^2}{5} + \frac{130^2}{5} + \frac{100^2}{5} - \frac{335^2}{15}. \end{aligned}$$

Compute each term:

$$\begin{aligned} \frac{105^2}{5} &= \frac{11025}{5} = 2205, \\ \frac{130^2}{5} &= \frac{16900}{5} = 3380, \\ \frac{100^2}{5} &= \frac{10000}{5} = 2000. \end{aligned}$$

So

$$\begin{aligned} \text{SSB} &= 2205 + 3380 + 2000 - 7489 \\ &= 7585 - 7489 \\ &= 96. \end{aligned}$$

Step 6: Calculate SSW

$$\begin{aligned} \text{SSW} &= \text{SST} - \text{SSB} \\ &= 126 - 96 \\ &= 30. \end{aligned}$$

Step 7: Mean squares Here $k = 3$, $N = 15$.

$$\begin{aligned} \text{MSB} &= \frac{\text{SSB}}{k - 1} = \frac{96}{2} = 48, \\ \text{MSW} &= \frac{\text{SSW}}{N - k} = \frac{30}{12} = 2.5. \end{aligned}$$

Step 8: F-statistic

$$F = \frac{\text{MSB}}{\text{MSW}} = \frac{48}{2.5} = 19.2.$$

Degrees of freedom:

$$\begin{aligned} v_1 &= k - 1 = 2, \\ v_2 &= N - k = 12. \end{aligned}$$

From F -tables at 5% LOS,

$$F_{0.05}(2, 12) = 3.89.$$

Since

$$F = 19.2 > 3.89,$$

we reject H_0 .

Final Answer

There is a significant difference in average sales between the three regions at the 5% level of significance.

Example 3: Completely Randomised Design (CRD)**Question**

A completely randomised design experiment with 10 plots and 3 treatments gave the following results:

Plot No.	1	2	3	4	5	6	7	8	9	10
Treatment	A	B	C	A	C	C	A	B	A	B
Yield	5	4	3	7	5	1	3	4	1	7

Analyse the results for treatment effects using one-way ANOVA.

Answer**Step 1: Rearranging by treatments**

Treatment	Yield from plots x_{ij}			
A	5	7	3	1
B	4	4	7	—
C	3	5	1	—

Thus

A: 5, 7, 3, 1,

B: 4, 4, 7,

C: 3, 5, 1.

Step 2: Totals and squares

Treatment totals:

$$T_A = 5 + 7 + 3 + 1 = 16,$$

$$T_B = 4 + 4 + 7 = 15,$$

$$T_C = 3 + 5 + 1 = 9.$$

Grand total:

$$T = T_A + T_B + T_C = 16 + 15 + 9 = 40.$$

Squares:

$$\sum x_{ij}^2 \text{ for A} = 5^2 + 7^2 + 3^2 + 1^2 = 25 + 49 + 9 + 1 = 84,$$

$$\sum x_{ij}^2 \text{ for B} = 4^2 + 4^2 + 7^2 = 16 + 16 + 49 = 81,$$

$$\sum x_{ij}^2 \text{ for C} = 3^2 + 5^2 + 1^2 = 9 + 25 + 1 = 35.$$

Hence

$$\sum \sum x_{ij}^2 = 84 + 81 + 35 = 200.$$

Total number of observations:

$$N = 10.$$

Step 3: SST

$$\begin{aligned} \text{SST} &= \sum \sum x_{ij}^2 - \frac{T^2}{N} \\ &= 200 - \frac{40^2}{10} \\ &= 200 - \frac{1600}{10} \\ &= 200 - 160 \\ &= 40. \end{aligned}$$

Step 4: SSB

Sample sizes:

$$n_A = 4, \quad n_B = 3, \quad n_C = 3.$$

$$\begin{aligned} \text{SSB} &= \frac{T_A^2}{n_A} + \frac{T_B^2}{n_B} + \frac{T_C^2}{n_C} - \frac{T^2}{N} \\ &= \frac{16^2}{4} + \frac{15^2}{3} + \frac{9^2}{3} - \frac{40^2}{10}. \end{aligned}$$

Compute:

$$\begin{aligned} \frac{16^2}{4} &= \frac{256}{4} = 64, \\ \frac{15^2}{3} &= \frac{225}{3} = 75, \\ \frac{9^2}{3} &= \frac{81}{3} = 27. \end{aligned}$$

So

$$\begin{aligned} \text{SSB} &= 64 + 75 + 27 - 160 \\ &= 166 - 160 \\ &= 6. \end{aligned}$$

Step 5: SSW

$$\begin{aligned} \text{SSW} &= \text{SST} - \text{SSB} \\ &= 40 - 6 = 34. \end{aligned}$$

Step 6: ANOVA table

Here $k = 3$, $N = 10$.

$$v_1 = k - 1 = 2,$$

$$v_2 = N - k = 7.$$

Mean squares:

$$\text{MSB} = \frac{\text{SSB}}{v_1} = \frac{6}{2} = 3,$$

$$\text{MSW} = \frac{\text{SSW}}{v_2} = \frac{34}{7} \approx 4.86.$$

$$F = \frac{\text{MSB}}{\text{MSW}} \approx \frac{3}{4.86} \approx 0.62.$$

Step 7: Decision

From F -tables at 5% LOS with $(2, 7)$ degrees of freedom,

$$F_{0.05}(2, 7) \approx 4.74.$$

Since

$$F \approx 0.62 < 4.74,$$

we fail to reject H_0 .

Final Answer

There is no significant difference between the treatment means at the 5% level of significance for this completely randomised design.

Example 4: Electric Lamps (Practice Problem)**Question**

The following table shows the lives (in hours) of four brands of electric lamps:

Brand A: 1610, 1610, 1650, 1680, 1700, 1720, 1800

Brand B: 1580, 1640, 1640, 1700, 1750

Brand C: 1460, 1550, 1600, 1620, 1640, 1660, 1740, 1820

Brand D: 1510, 1520, 1530, 1570, 1600, 1680

Perform an analysis of variance and test the homogeneity of the mean lives of the four brands of lamps at 5% LOS.

Introduction to Two-Factor ANOVA

Definition

In a **two-factor ANOVA**, we analyse the impact of two categorical independent variables (factors) on a dependent variable.

This method is used to:

- determine whether each factor has a significant effect on the response variable, and
- study how the variation in the data can be attributed to the factors and to random error.

Typically we consider:

- Factor A with a levels,
- Factor B with b levels,
- one observation x_{ij} at each combination (i, j) (no replication).

Components of Two-Factor ANOVA

Definition

1. Total Sum of Squares (SST)

Measures the total variation in the data:

$$SST = \sum_{i=1}^a \sum_{j=1}^b x_{ij}^2 - \frac{T^2}{N},$$

where x_{ij} is the individual observation, T is the total of all observations, and $N = ab$ is the total number of observations.

Definition

2. Sum of Squares for Factor A (SSA)

Measures the variation due to differences between levels of Factor A:

$$SSA = \frac{1}{b} \sum_{i=1}^a T_{A_i}^2 - \frac{T^2}{N},$$

where T_{A_i} is the total for the i -th level of Factor A and b is the number of levels of Factor B.

Definition

3. Sum of Squares for Factor B (SSB)

Measures the variation due to differences between levels of Factor B:

$$SSB = \frac{1}{a} \sum_{j=1}^b T_{B_j}^2 - \frac{T^2}{N},$$

where T_{B_j} is the total for the j -th level of Factor B and a is the number of levels of Factor A.

Definition

4. Sum of Squares for Error (SSE)

Measures the random variation within the data that cannot be explained by the factors:

$$SSE = SST - SSA - SSB.$$

Mean Squares and F-Statistics

Definition

Mean Squares

$$\begin{aligned} MSA &= \frac{SSA}{a-1} && \text{(Mean Square for Factor A),} \\ MSB &= \frac{SSB}{b-1} && \text{(Mean Square for Factor B),} \\ MSE &= \frac{SSE}{(a-1)(b-1)} && \text{(Mean Square for Error).} \end{aligned}$$

Definition

F-Statistics

$$\begin{aligned} F_A &= \frac{MSA}{MSE} && \text{with } (v_1, v_2) = (a-1, (a-1)(b-1)) \text{ d.f.,} \\ F_B &= \frac{MSB}{MSE} && \text{with } (v_1, v_2) = (b-1, (a-1)(b-1)) \text{ d.f..} \end{aligned}$$

If F_A or F_B exceeds the corresponding critical F value at the chosen level of significance, we reject H_0 for that factor.

Worked Example 1: Crop Variety and Soil Type

Question

Problem. Suppose two varieties of crops (A and B) are tested in three different soil types (1, 2, and 3). The yield (in kg) for each combination of crop variety and soil type is recorded in a randomised experiment.

Soil Type	Crop A	Crop B
1	10	12
2	9	11
3	13	14

We want to determine whether there is a significant difference in yield due to crop variety or soil type using two-factor ANOVA (no replication).

Answer

Let Factor A = *Crop variety* (A, B) $\Rightarrow a = 2$, and Factor B = *Soil type* (1,2,3) $\Rightarrow b = 3$. Total observations $N = ab = 6$.

1. Calculate the totals

$$T_A = 10 + 9 + 13 = 32,$$

$$T_B = 12 + 11 + 14 = 37,$$

$$T_1 = 10 + 12 = 22,$$

$$T_2 = 9 + 11 = 20,$$

$$T_3 = 13 + 14 = 27,$$

$$T = 32 + 37 = 69.$$

2. Calculate SST

$$\begin{aligned}
 \text{SST} &= \sum_i \sum_j x_{ij}^2 - \frac{T^2}{N} \\
 &= (10^2 + 9^2 + 13^2 + 12^2 + 11^2 + 14^2) - \frac{69^2}{6} \\
 &= (100 + 81 + 169 + 144 + 121 + 196) - \frac{4761}{6} \\
 &= 811 - 793.5 \\
 &= 17.5.
 \end{aligned}$$

3. Calculate SSA (Factor A – crop variety)

$$\begin{aligned}
SSA &= \frac{1}{b} \sum_{i=1}^a T_{A_i}^2 - \frac{T^2}{N} \\
&= \frac{1}{3}(32^2 + 37^2) - 793.5 \\
&= \frac{1}{3}(1024 + 1369) - 793.5 \\
&= \frac{2393}{3} - 793.5 \\
&\approx 797.67 - 793.5 \\
&\approx 4.17.
\end{aligned}$$

4. Calculate SSB (Factor B – soil type)

$$\begin{aligned}
SSB &= \frac{1}{a} \sum_{j=1}^b T_{B_j}^2 - \frac{T^2}{N} \\
&= \frac{1}{2}(22^2 + 20^2 + 27^2) - 793.5 \\
&= \frac{1}{2}(484 + 400 + 729) - 793.5 \\
&= \frac{1613}{2} - 793.5 \\
&= 806.5 - 793.5 \\
&= 13.
\end{aligned}$$

5. Calculate SSE (Error)

$$\begin{aligned}
SSE &= SST - SSA - SSB \\
&= 17.5 - 4.17 - 13 \\
&\approx 0.33.
\end{aligned}$$

6. Calculate Mean Squares Degrees of freedom:

$$\begin{aligned}
v_A &= a - 1 = 1, \\
v_B &= b - 1 = 2, \\
v_E &= (a - 1)(b - 1) = 1 \times 2 = 2.
\end{aligned}$$

$$MSA = \frac{SSA}{a - 1} = \frac{4.17}{1} = 4.17,$$

$$MSB = \frac{SSB}{b - 1} = \frac{13}{2} = 6.5,$$

$$MSE = \frac{SSE}{(a - 1)(b - 1)} = \frac{0.33}{2} \approx 0.165.$$

7. F-statistics

$$F_A = \frac{MSA}{MSE} = \frac{4.17}{0.165} \approx 25.27,$$

$$F_B = \frac{MSB}{MSE} = \frac{6.5}{0.165} \approx 39.39.$$

At $\alpha = 0.05$:

- For Factor A, compare F_A with $F_{0.05}(1, 2) \approx 18.51$.
- For Factor B, compare F_B with $F_{0.05}(2, 2) \approx 19$.

Since $F_A > 18.51$ and $F_B > 19$, we reject H_0 for both factors.

Final Answer

There is a significant difference in yield with respect to both the type of crop and the type of soil.

Practice Problem: Machines and Shifts**Question**

A factory wants to test if two machines (Machine A and Machine B) have different production times (in hours) during three shifts (Morning, Afternoon, and Night). The data are as follows:

Shift	Machine A	Machine B
Morning	8	10
Afternoon	9	8
Night	7	11

Perform a two-factor ANOVA (Machine as Factor A, Shift as Factor B) and comment on the significance of machine and shift effects.

Example 2: Randomised Block Design (Three Crops)**Question**

Example. Three varieties of a crop are tested in a randomised block design with four replications (blocks). The yields (in kilograms) are:

Crops

Blocks	A	B	C
1	47	49	48
2	51	49	53
3	49	52	52
4	49	50	51

Analyse the data for significance of crop varieties and block effects using two-way ANOVA (treatments and blocks).

Answer

Interpret this as a two-factor ANOVA where:

- Factor A (Treatments): Crop varieties A, B, C ($a = 3$),
- Factor B (Blocks): 1, 2, 3, 4 ($b = 4$).

Total observations: $N = ab = 12$.

1. Treatment and block totals Treatment totals

$$T_A = 47 + 51 + 49 + 49 = 196,$$

$$T_B = 49 + 49 + 52 + 50 = 200,$$

$$T_C = 48 + 53 + 52 + 51 = 204.$$

Block totals

$$B_1 = 47 + 49 + 48 = 144,$$

$$B_2 = 51 + 49 + 53 = 153,$$

$$B_3 = 49 + 52 + 52 = 153,$$

$$B_4 = 49 + 50 + 51 = 150.$$

Grand total

$$T = 196 + 200 + 204 = 600.$$

2. Sum of squares Total sum of squares (SST)

Compute $\sum \sum x_{ij}^2$:

$$\sum \sum x_{ij}^2 = 30036 \quad (\text{from all 12 squared yields}),$$

$$\frac{T^2}{N} = \frac{600^2}{12} = \frac{360000}{12} = 30000.$$

Hence

$$\text{SST} = 30036 - 30000 = 36.$$

Sum of squares for treatments (crops)

$$\begin{aligned}
\sum T_i^2 &= 196^2 + 200^2 + 204^2 \\
&= 38416 + 40000 + 41616 \\
&= 120032.
\end{aligned}$$

With $b = 4$ blocks,

$$\begin{aligned}
\text{SSTreat} &= \frac{1}{b} \sum T_i^2 - \frac{T^2}{N} \\
&= \frac{1}{4}(120032) - 30000 \\
&= 30008 - 30000 \\
&= 8.
\end{aligned}$$

Sum of squares for blocks

$$\begin{aligned}
\sum B_j^2 &= 144^2 + 153^2 + 153^2 + 150^2 \\
&= 20736 + 23409 + 23409 + 22500 \\
&= 90054.
\end{aligned}$$

With $a = 3$ treatments,

$$\begin{aligned}
\text{SSBlock} &= \frac{1}{a} \sum B_j^2 - \frac{T^2}{N} \\
&= \frac{1}{3}(90054) - 30000 \\
&= 30018 - 30000 \\
&= 18.
\end{aligned}$$

Error sum of squares

$$\begin{aligned}
\text{SSE} &= \text{SST} - \text{SSTreat} - \text{SSBlock} \\
&= 36 - 8 - 18 \\
&= 10.
\end{aligned}$$

3. Degrees of freedom and mean squares

$$\begin{aligned}
v_{\text{Total}} &= N - 1 = 11, \\
v_{\text{Treat}} &= a - 1 = 2, \\
v_{\text{Block}} &= b - 1 = 3, \\
v_{\text{Error}} &= (a - 1)(b - 1) = 2 \times 3 = 6.
\end{aligned}$$

Mean squares:

$$\begin{aligned} \text{MSTreat} &= \frac{\text{SSTreat}}{v_{\text{Treat}}} = \frac{8}{2} = 4, \\ \text{MSBlock} &= \frac{\text{SSBlock}}{v_{\text{Block}}} = \frac{18}{3} = 6, \\ \text{MSE} &= \frac{\text{SSE}}{v_{\text{Error}}} = \frac{10}{6} \approx 1.67. \end{aligned}$$

4. F-statistics

$$\begin{aligned} F_{\text{Treat}} &= \frac{\text{MSTreat}}{\text{MSE}} \approx \frac{4}{1.67} \approx 2.40, \\ F_{\text{Block}} &= \frac{\text{MSBlock}}{\text{MSE}} \approx \frac{6}{1.67} \approx 3.60. \end{aligned}$$

5. Decision At $\alpha = 0.05$:

- For treatments: compare F_{Treat} with $F_{0.05}(2, 6)$.
- For blocks: compare F_{Block} with $F_{0.05}(3, 6)$.

From standard F -tables, $F_{0.05}(2, 6)$ and $F_{0.05}(3, 6)$ are both larger than the observed values 2.40 and 3.60, respectively.

Hence both F_{Treat} and F_{Block} are *not* significant at the 5% level.

Final Answer

At the 5% level of significance, there is no evidence of a significant difference between the three crop varieties, nor is there a significant block (field) effect in this randomised block design.

Introduction to Latin Square Design

Definition

A **Latin Square Design** is a type of experimental design used when there are two extraneous sources of variability (aside from the treatment or factor of interest) that need to be controlled. It is commonly applied in agricultural, industrial, or laboratory experiments when we want to test treatments while controlling for both row and column effects.

Key Components:

- **Treatments (T):** Primary factors under study.
- **Rows (R):** One blocking factor (e.g. plots, days, locations).
- **Columns (C):** Second blocking factor, independent of rows (e.g. soil types, machines, engines).

Each treatment appears exactly once in every row and once in every column.

Statistical Formulas in Latin Square Design

Definition

Notation

- n = order of the Latin square (number of rows/columns/treatments).
- x_{ij} = observation in row i , column j .
- R_i = total of observations in row i .
- C_j = total of observations in column j .
- T_k = total of observations for treatment k .
- T = grand total of all observations.
- $N = n^2$ = total number of observations.

Definition

1. Total Sum of Squares (SST)

$$\text{SST} = \sum_{i=1}^n \sum_{j=1}^n x_{ij}^2 - \frac{T^2}{N}.$$

T is the grand total and N is the total number of observations.

Definition

2. Sum of Squares for Rows (SSA)

$$\text{SSA} = \frac{1}{n} \sum_{i=1}^n R_i^2 - \frac{T^2}{N},$$

where R_i is the total of row i .

Definition

3. Sum of Squares for Columns (SSB)

$$\text{SSB} = \frac{1}{n} \sum_{j=1}^n C_j^2 - \frac{T^2}{N},$$

where C_j is the total of column j .

Definition**4. Sum of Squares for Treatments (SSL)**

$$SSL = \frac{1}{n} \sum_{k=1}^n T_k^2 - \frac{T^2}{N},$$

where T_k is the total for treatment k .

Definition**5. Sum of Squares for Error (SSE)**

$$SSE = SST - SSA - SSB - SSL.$$

SSE represents unexplained variation after accounting for rows, columns, and treatments.

Mean Squares and F-Statistics**Definition****Mean Squares**

$MSA = \frac{SSA}{n-1}$	(Mean Square for Rows),
$MSB = \frac{SSB}{n-1}$	(Mean Square for Columns),
$MSL = \frac{SSL}{n-1}$	(Mean Square for Treatments),
$MSE = \frac{SSE}{(n-1)(n-2)}$	(Mean Square for Error).

Definition**F-Statistics and Degrees of Freedom**

$F_A = \frac{MSA}{MSE}$	for Rows, with DoF $(n-1, (n-1)(n-2))$,
$F_B = \frac{MSB}{MSE}$	for Columns, with DoF $(n-1, (n-1)(n-2))$,
$F_L = \frac{MSL}{MSE}$	for Treatments, with DoF $(n-1, (n-1)(n-2))$.

Compare each F with the critical value from the F -table at the chosen level of significance. If F exceeds the critical value, the corresponding effect is significant.

Problem 1: Yield of Crops under Different Fertilizers

Question

Yield of Crops under Different Fertilizers (3×3 Latin Square).

You are testing three types of fertilizers (A, B, C) on crop yield, with three plots (rows) and three soil types (columns).

	Soil 1	Soil 2	Soil 3
Plot 1	A: 6	B: 7	C: 8
Plot 2	B: 8	C: 9	A: 7
Plot 3	C: 5	A: 6	B: 8

Test whether there are significant differences between plots (rows), soil types (columns), and fertilizers (treatments) at the 5% LOS.

Answer

Here $n = 3$ (a 3×3 Latin square), so $N = 9$.

1. Observed yields

	Soil 1	Soil 2	Soil 3
Plot 1	6	7	8
Plot 2	8	9	7
Plot 3	5	6	8

2. Row totals

$$R_1 = 6 + 7 + 8 = 21,$$

$$R_2 = 8 + 9 + 7 = 24,$$

$$R_3 = 5 + 6 + 8 = 19.$$

3. Column totals

$$C_1 = 6 + 8 + 5 = 19,$$

$$C_2 = 7 + 9 + 6 = 22,$$

$$C_3 = 8 + 7 + 8 = 23.$$

4. Treatment totals

$$T_A = 6 + 7 + 6 = 19,$$

$$T_B = 7 + 8 + 8 = 23,$$

$$T_C = 8 + 9 + 5 = 22.$$

5. Grand total and number of observations

$$T = 6 + 7 + 8 + 8 + 9 + 7 + 5 + 6 + 8 = 64,$$

$$N = 9.$$

6. Total sum of squares (SST)

$$\begin{aligned}
\sum x_{ij}^2 &= 6^2 + 7^2 + 8^2 + 8^2 + 9^2 + 7^2 + 5^2 + 6^2 + 8^2 \\
&= 468, \\
\frac{T^2}{N} &= \frac{64^2}{9} = \frac{4096}{9} \approx 455.11.
\end{aligned}$$

$$\text{SST} = 468 - 455.11 \approx 12.89.$$

7. Sum of squares for rows (SSA)

$$\begin{aligned}
\text{SSA} &= \frac{1}{n} (R_1^2 + R_2^2 + R_3^2) - \frac{T^2}{N} \\
&= \frac{1}{3} (21^2 + 24^2 + 19^2) - \frac{64^2}{9} \\
&= \frac{1}{3} (441 + 576 + 361) - 455.11 \\
&= 459.33 - 455.11 \approx 4.22.
\end{aligned}$$

8. Sum of squares for columns (SSB)

$$\begin{aligned}
\text{SSB} &= \frac{1}{n} (C_1^2 + C_2^2 + C_3^2) - \frac{T^2}{N} \\
&= \frac{1}{3} (19^2 + 22^2 + 23^2) - 455.11 \\
&= \frac{1}{3} (361 + 484 + 529) - 455.11 \\
&= 458.00 - 455.11 \approx 2.89.
\end{aligned}$$

9. Sum of squares for treatments (SSL)

$$\begin{aligned}
\text{SSL} &= \frac{1}{n} (T_A^2 + T_B^2 + T_C^2) - \frac{T^2}{N} \\
&= \frac{1}{3} (19^2 + 23^2 + 22^2) - 455.11 \\
&= \frac{1}{3} (361 + 529 + 484) - 455.11 \\
&= 458.00 - 455.11 \approx 2.89.
\end{aligned}$$

10. Error sum of squares (SSE)

$$\begin{aligned}
\text{SSE} &= \text{SST} - \text{SSA} - \text{SSB} - \text{SSL} \\
&\approx 12.89 - 4.22 - 2.89 - 2.89 \\
&\approx 2.89.
\end{aligned}$$

11. Mean squares Since $n = 3$:

$$MSA = \frac{SSA}{n-1} = \frac{4.22}{2} \approx 2.11,$$

$$MSB = \frac{SSB}{n-1} = \frac{2.89}{2} \approx 1.445,$$

$$MSL = \frac{SSL}{n-1} = \frac{2.89}{2} \approx 1.445,$$

$$MSE = \frac{SSE}{(n-1)(n-2)} = \frac{2.89}{2 \cdot 1} \approx 1.445.$$

12. F-statistics

$$F_A = \frac{MSA}{MSE} \approx \frac{2.11}{1.445} \approx 1.46,$$

$$F_B = \frac{MSB}{MSE} \approx \frac{1.445}{1.445} \approx 1.00,$$

$$F_L = \frac{MSL}{MSE} \approx \frac{1.445}{1.445} \approx 1.00.$$

For DoF (2, 2) and $\alpha = 0.05$:

$$F_{0.05}(2, 2) = 19.0.$$

Since all calculated F -values are far less than 19.0, none are significant at the 5% level.

Final Answer

There is no significant difference in yield between rows (plots), columns (soil types), or fertilizers in this 3×3 Latin square experiment at the 5% LOS.

Example 2: Comparison of Burners

Question

The following data resulted from an experiment to compare three burners B_1 , B_2 and B_3 . A Latin square design was used as the tests were made on three engines and were spread over three days.

	Engine 1	Engine 2	Engine 3
Day 1	$B_1 : 16$	$B_2 : 17$	$B_3 : 20$
Day 2	$B_2 : 16$	$B_3 : 21$	$B_1 : 15$
Day 3	$B_3 : 15$	$B_1 : 12$	$B_2 : 13$

Test the hypothesis that there is no difference between the burners.

Answer

This is a 3×3 Latin square ($n = 3, N = 9$).

1. Row (day) totals

$$R_1 = 16 + 17 + 20 = 53,$$

$$R_2 = 16 + 21 + 15 = 52,$$

$$R_3 = 15 + 12 + 13 = 40.$$

2. Column (engine) totals

$$C_1 = 16 + 16 + 15 = 47,$$

$$C_2 = 17 + 21 + 12 = 50,$$

$$C_3 = 20 + 15 + 13 = 48.$$

3. Treatment (burner) totals

$$T_{B_1} = 16 + 15 + 12 = 43,$$

$$T_{B_2} = 17 + 16 + 13 = 46,$$

$$T_{B_3} = 20 + 21 + 15 = 56.$$

4. Grand total

$$T = 43 + 46 + 56 = 145, \quad N = 9.$$

5. Total sum of squares (SST)

$$\begin{aligned} \sum x_{ij}^2 &= 16^2 + 17^2 + 20^2 + 16^2 + 21^2 + 15^2 + 15^2 + 12^2 + 13^2 \\ &= 2405, \end{aligned}$$

$$\frac{T^2}{N} = \frac{145^2}{9} = \frac{21025}{9} \approx 2336.11.$$

$$SST = 2405 - 2336.11 \approx 68.89.$$

6. Sum of squares for rows (SSA)

$$\begin{aligned} SSA &= \frac{1}{n} \sum R_i^2 - \frac{T^2}{N} \\ &= \frac{1}{3}(53^2 + 52^2 + 40^2) - \frac{145^2}{9} \\ &= \frac{1}{3}(2809 + 2704 + 1600) - 2336.11 \\ &= 2371.00 - 2336.11 \approx 34.89. \end{aligned}$$

7. Sum of squares for columns (SSB)

$$\begin{aligned}
SSB &= \frac{1}{n} \sum C_j^2 - \frac{T^2}{N} \\
&= \frac{1}{3}(47^2 + 50^2 + 48^2) - 2336.11 \\
&= \frac{1}{3}(2209 + 2500 + 2304) - 2336.11 \\
&= 2337.67 - 2336.11 \approx 1.56.
\end{aligned}$$

8. Sum of squares for treatments (SSL)

$$\begin{aligned}
SSL &= \frac{1}{n} \sum T_k^2 - \frac{T^2}{N} \\
&= \frac{1}{3}(43^2 + 46^2 + 56^2) - 2336.11 \\
&= \frac{1}{3}(1849 + 2116 + 3136) - 2336.11 \\
&= 2367.00 - 2336.11 \approx 30.89.
\end{aligned}$$

9. Error sum of squares (SSE)

$$\begin{aligned}
SSE &= SST - SSA - SSB - SSL \\
&\approx 68.89 - 34.89 - 1.56 - 30.89 \\
&\approx 1.55.
\end{aligned}$$

10. Mean squares and F-statistics With $n = 3$,

$$\begin{aligned}
MSA &= \frac{SSA}{n-1} \approx \frac{34.89}{2} \approx 17.45, \\
MSB &= \frac{SSB}{n-1} \approx \frac{1.56}{2} \approx 0.78, \\
MSL &= \frac{SSL}{n-1} \approx \frac{30.89}{2} \approx 15.45, \\
MSE &= \frac{SSE}{(n-1)(n-2)} \approx \frac{1.55}{2} \approx 0.78.
\end{aligned}$$

Hence

$$\begin{aligned}
F_A &\approx \frac{17.45}{0.78} \approx 22.5, \\
F_B &\approx \frac{0.78}{0.78} \approx 1.0, \\
F_L &\approx \frac{15.45}{0.78} \approx 19.9.
\end{aligned}$$

For DoF (2, 2) at $\alpha = 0.05$,

$$F_{0.05}(2, 2) = 19.0.$$

Therefore:

- $F_A \approx 22.5 > 19.0 \Rightarrow$ row (day) effect is significant.
- $F_B \approx 1.0 < 19.0 \Rightarrow$ column (engine) effect is not significant.
- $F_L \approx 19.9 > 19.0 \Rightarrow$ burner (treatment) effect is significant.

Final Answer

There is a significant difference between the burners, and also a significant day effect. Engine differences are not significant. Thus the three burners do not have the same mean performance.

Example 3: Methods of Cultivation (Practice)

Question

Analyse the variance in the following Latin square of yields (in kgs) of paddy, where A, B, C, and D denote four different methods of cultivation.

	Column 1	Column 2	Column 3	Column 4
Row 1	D:122	A:121	C:123	B:122
Row 2	B:124	C:123	A:122	D:125
Row 3	A:120	B:119	D:120	C:121
Row 4	C:122	D:123	B:121	A:122

Examine whether the different methods of cultivation have given significantly different yields, using the Latin square ANOVA procedure.

ANOVA Question Bank

Question

1. Four doctors each test four treatments for a certain disease and observe the number of days each patient takes to recover. The results are as follows (recovery time in days):

Doctor	Treatment 1	Treatment 2	Treatment 3	Treatment 4
A	10	14	19	20
B	11	15	17	21
C	9	12	16	19
D	8	13	17	20

Discuss the difference between:

- doctors
- treatments.

Answer**Question**

2. The following data represent the number of units of production per day turned out by five different workers using four different types of machines: :contentReference[oaicite:2]index=2

Workers	Machine A	Machine B	Machine C	Machine D
1	44	38	47	36
2	46	40	52	43
3	34	36	44	32
4	43	38	46	33
5	38	42	49	39

- (a) Test whether the five workers differ with respect to mean productivity.
(b) Test whether the mean productivity is the same for the four different machine types.

Answer

Question

3. In order to determine whether there is a significant difference in the durability of three makes of computers, samples of size five are selected from each make and the frequency of repairs during the first year of purchase is observed. The results are:

	Make A	Make B	Make C
	5	8	7
	6	10	3
	8	11	5
	9	12	4
	7	4	1

State the conclusion based on ANOVA.

Answer**Question**

4. A varietal trial was conducted on wheat with four varieties A, B, C, D in a Latin square design. The plan and observed yield per plot are given:

	Col 1	Col 2	Col 3	Col 4
Row 1	C: 25	B: 23	A: 20	D: 20
Row 2	A: 19	D: 19	C: 21	B: 18
Row 3	B: 19	A: 14	D: 17	C: 20
Row 4	D: 17	C: 20	B: 21	A: 15

Analyse the data and interpret the results.

Answer